Md Ashiqur Rahman

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Education

Purdue University **West Lafayette, IN**
Doctor of Philosophy in Computer Science *2021–*
GPA – 3.96/4.0

Advisor: Raymond A. Yeh

Awarded Bilsland Dissertation Fellowship

Purdue University **West Lafayette, IN**
Master of Science in Computer Science *2021-2025*

Bangladesh University of Engg & Tech **Dhaka, Bangladesh**
Bachelor of Science in Computer Science and Engineering *2015–2019*
GPA – 3.93/4.0

Advisor: Md. Shamsuzzoha Bayzid

Professional Employment

Purdue University **West Lafayette, IN**
Graduate Assistant in the Department of Computer Science *2021-present*

- Teaching Assistant for courses on deep learning and computer graphics.

Autodesk **San Francisco, CA**
Research Intern *Summer 2025*
Manager: Robert Giaquint

- Researched CAD model generation with large vision language models (VLMs).

NVIDIA **Santa Clara, CA**
Research Intern *Summer 2023*
Manager: Anima Anandkumar

- Researched large-scale weather prediction and foundation model for scientific computing.

United International University **Dhaka Bangladesh**
Lecturer in the Department of Computer Science *2019-2021*

- Taught courses on computer graphics, networking systems, and data structures.

Publications

Refereed Conference

[C1] **Md Ashiqur Rahman**, Jun Hao Lim, Jeremiah Jiang, Teck-Yian Lim, and Raymond A. Yeh. Tunable soft equivariance with guarantees. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (*CVPR*). IEEE/CVF, 2026.

[C2] **Md Ashiqur Rahman**, Chiao-An Yang, Michael N. Cheng, Jun Hao Lim, Jeremiah Jiang, Teck-Yian Lim, and Raymond A. Yeh. Local scale equivariance with latent deep equilibrium canonicalizer. In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (*ICCV*),

2025.

[C3] Tinghan Yang, **Md Ashiqur Rahman**, and Raymond A. Yeh. Clipsym: Delving into symmetry detection with clip. In *International Conference on Computer Vision* (*ICCV*), 2025.

[C4] **Md Ashiqur Rahman** and Raymond A. Yeh. Group downsampling with equivariant anti-aliasing. In *International Conference on Learning Representations* (*ICLR*), 2025.

[C5] **Md Ashiqur Rahman**, R. J. George, M. Elleithy, D. Leibovici, Z. Li, B. Bonev, C. White, J. Berner, R. A. Yeh, J. Kossaifi, K. Azizzadenesheli, and A. Anandkumar. Pretraining codomain attention neural operators for solving multiphysics PDEs. In *Advances in Neural Information Processing Systems* (*NeurIPS*), 2024.

[C6] **Md Ashiqur Rahman** and Raymond A. Yeh. Truly scale-equivariant deep nets with fourier layers. In *Advances in Neural Information Processing Systems* (*NeurIPS*), 2023.

[C7] **Md Ashiqur Rahman**, Abdullah Aman Tutul, and A. B. M. Alim Al Islam. Solving the maze of diagnosing parkinson's disease based on portable eeg sensing to be adaptable to go in-the-wild. *Proceedings of the 7th International Conference on Networking, Systems and Security*, 2020. (Best Paper Award).

[C8] Akm Ashiquzzaman, Abdul Kawsar Tushar, **Md Ashiqur Rahman**, and Farzana Mohsin. An efficient recognition method for handwritten arabic numerals using cnn with data augmentation and dropout. *Data Management, Analytics and Innovation*, 2018.

Refereed Journal

[J1] **Md Ashiqur Rahman**, Abdullah Aman Tutul, Mahfuza Sharmin, and Md. Shamsuzzoha Bayzid. Beene: deep learning-based nonlinear embedding improves batch effect estimation. *Bioinformatics*, 2023.

[J2] **Md Ashiqur Rahman**, Manuel A. Florez, Anima Anandkumar, Zachary E. Ross, and Kamyar Azizzadenesheli. Generative adversarial neural operators. *Transactions on Machine Learning Research*, 2022.

[J3] **Md Ashiqur Rahman**, Zachary E. Ross, and Kamyar Azizzadenesheli. U-no: U-shaped neural operators. *Transactions on Machine Learning Research*, 2022.

[J4] **Md Ashiqur Rahman**, Abdullah Aman Tutul, Sifat Muhammad Abdullah, and Md. Shamsuzzoha Bayzid. Chapao: Likelihood and hierarchical reference-based representation of biomolecular sequences and applications to compressing multiple sequence alignments. *PLoS ONE*, 2022.

Preprints / In Submission

[S1] Hrishikesh Viswanath, Md Ashiqur Rahman, Chi Lin, Damon Conover, and Aniket Bera. Hessianforge: Scalable lidar reconstruction with physics-informed neural representation and smoothness energy constraints. *arXiv:2503.08929*, 2025.

[S2] Hrishikesh Viswanath, **Md Ashiqur Rahman**, Abhijeet Vyas, Andrey Shor, Beatriz Medeiros, Stephanie Hernandez, Suhas Eswarappa Prameela, and Aniket Bera. Neural operator: Is data all you need to model the world? an insight into the impact of physics informed machine learning. 2023.

[S3] **Md Ashiqur Rahman**, Jasorsi Ghosh, Hrishikesh Viswanath, Kamyar Azizzadenesheli, and Aniket Bera. Pacmo: Partner dependent human motion generation in dyadic human activity using neural operators. *ArXiv*, abs/2211.16210, 2022.

[S4] Hrishikesh Viswanath, **Md Ashiqur Rahman**, Rashmi Bhaskara, and Aniket Bera. Adafnio: Adaptive fourier neural interpolation operator for video frame interpolation. 2022.

Awards & Recognition

Bilsland Dissertation Fellowship *2026-2027*

- Awarded by Purdue University Graduate School.

ACCESS Discover Project Award *2025*

- Awarded 150,000 computing credits (∼$10K–15K equivalent) in national cyberinfrastructure resources to support scientific discovery.

Teaching Experience

Purdue University **West Lafayette, IN**
Teaching Assistant - CS587: Foundations of Deep Learning *Spring 2024*

- Co-designed and graded course assignments and exam questions involving equivariant models, optimizing deep neural networks, generative models, and optimization layers.

Purdue University **West Lafayette, IN**
Teaching Assistant - CS373: Data Mining and Machine Learning *Fall 2022, Spring 2023*

- Co-designed course assignments and exam questions involving fundamentals of machine learning, perceptron, SVM, learning theory, and data privacy.

Purdue University **West Lafayette, IN**
Teaching Assistant - CS334: Fundamentals of Computer Graphics *Fall 2021, Fall 2023*

- Co-designed course assignments and exam questions involving GPU programming, ray tracing, texture mapping, and procedural modeling.

United International University **Dhaka, Bangladesh.**
Lecturer *2019 - 2021*

- Redesigned and instructed courses, including Data Structures and Algorithms, Simulation and Modeling, Intro to Computer Graphics, and Computer Networking.

Services

Professional Service

Conference Reviewer:

- Neural Information Processing Systems, 2022 – 2025
- International Conference on Machine Learning, 2023 – 2025
- International Conference on Learning Representations, 2023 – 2025
- Computer Vision and Pattern Recognition, 2024

Journal Reviewer:

- IET Computer Vision
- IEEE Transactions on Neural Networks and Learning Systems

Research with Undergraduates

Michael Cheng
Undergraduate, Purdue, Fall 2024 – Fall 2025. Conducting undergraduate research under the supervision of Professor Raymond A. Yeh. As a mentor, I receive weekly updates, provide feedback, set project directions, and offer hands-on guidance in project development and implementation.
Research topic: Equivariant machine learning for computer vision.

Changxiang Gao
Undergraduate, Purdue, Summer 2024 – Fall 2024. Conducted undergraduate research under the supervision of Professor Raymond A. Yeh. As a mentor, I received weekly updates, defined project goals, and provided hands-on guidance for implementation.
Research topic: Any-resolution image classification.

Anugu Arun Reddy
Visiting Scholar, Purdue, Summer 2024 under the supervision of Professor Aniket Bera. As a mentor, I defined the project scope, received weekly updates, and guided the implementation process.
Research topic: Human motion generation.